Questions

The following questions are meant to solidify key points to remember as well as tie the content back to your own experience:

1. What are some low-code options for data exploration that we discussed in this chapter?
2. When might you use Databricks Assistant for data exploration, and when might you use AutoML's data profile notebook?
3. How and why would you set expectations on your data?
4. When would you use a regular database versus a vector database? What are some common use cases for vector databases?

Answers

After putting thought into the questions, compare your answers to ours:

1. Some low-code data exploration options include using the `ydata` library, in-cell data profile, Databricks Assistant, and AutoML.
2. Databricks Assistant is useful for data exploration when you have a good idea of the analyses you want to build and you want code assistance. Databricks Assistant is a great way to speed up the coding process or augment your SQL knowledge. On the other hand, AutoML is very useful for automatically creating a profile notebook that broadly covers your dataset.
3. We would use Delta Live Tables to set expectations. Expectations are a way to flexibly handle data abnormalities and give the options to report bad data, drop that data, or fail the pipeline entirely.
4. Regular databases, or relational databases, are designed for data in tabular form, typically organized in rows or columns. A vector database is designed to store vector data, such as embeddings and high-dimensional data. Vector databases are optimized for operations based on vector space models, similarity searches, image and video analysis, and other ML problems. Some common use cases include **Retrieval Augmented Generation** (**RAG**) systems, recommendation systems, and image and video recognition.